US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12406973
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lin; Da-Jun et al.

Structure with photodiode, high electron mobility transistor, surface acoustic wave device and fabricating method of the same

Abstract

A structure with a photodiode, an HEMT and an SAW device includes a photodiode and an HEMT. The photodiode includes a first electrode and a second electrode. The first electrode contacts a P-type III-V semiconductor layer. The second electrode contacts an N-type III-V semiconductor layer. The HEMT includes a P-type gate disposed on an active layer. A gate electrode is disposed on the P-type gate. Two source/drain electrodes are respectively disposed at two sides of the P-type gate. Schottky contact is between the first electrode and the P-type III-V semiconductor layer, and between the gate electrode and the P-type gate. Ohmic contact is between the second electrode and the first N-type III-V semiconductor layer, and between one of the two source/drain electrodes and the active layer and between the other one of two source/drain electrodes and the active layer.

Inventors: **Lin; Da-Jun (Kaohsiung, TW), Chang; Chih-Wei (Tainan, TW), Tsai; Fu-Yu (Tainan, TW), Tsai; Bin-Siang (Changhua County, TW), Chiu; Chung-Yi (Tainan, TW)**

Applicant: **UNITED MICROELECTRONICS CORP.** (Hsin-Chu, TW)

Family ID: **90956877**

Assignee: **UNITED MICROELECTRONICS CORP. (Hsin-Chu, TW)**

Appl. No.: **18/077192**

Filed: **December 07, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240162208 A1	May. 16, 2024

Foreign Application Priority Data

CN	202211404042.2	Nov. 10, 2022

Publication Classification

Int. Cl.: **H10D30/01** (20250101); **H01L25/16** (20230101); **H03H3/08** (20060101); **H03H9/02** (20060101); **H10D30/47** (20250101); **H10D30/67** (20250101); **H10D62/85** (20250101); **H10D64/01** (20250101); **H10D64/23** (20250101); **H10D64/64** (20250101); **H10F71/00** (20250101); **H10F77/124** (20250101); **H10F77/14** (20250101); **H10F77/20** (20250101)

U.S. Cl.:

CPC **H01L25/167** (20130101); **H03H3/08** (20130101); **H03H9/02976** (20130101); **H10D30/015** (20250101); **H10D30/475** (20250101); **H10D30/6737** (20250101); **H10D30/6738** (20250101); **H10D30/675** (20250101); **H10D62/85** (20250101); **H10D62/8503** (20250101); **H10D64/01** (20250101); **H10D64/258** (20250101); **H10D64/64** (20250101); **H10F71/1278** (20250101); **H10F77/1246** (20250101); **H10F77/146** (20250101); **H10F77/206** (20250101);

Field of Classification Search

CPC: H10F (39/8037); H10F (39/199); H10F (30/21); H10F (30/221); H10F (30/2218); H10F (30/26); H10F (39/802); H10F (39/8057); H10F (39/8067); H10F (77/40); H10F (30/288); H10F (30/289); H10F (77/12-1285); H10F (71/00-1395); H10F (77/933); H10F (77/957-959); H10F (77/50); H10F (77/496); H10F (77/413); H10F (77/407); H10F (77/703); H10F (77/707); H10F (39/18-1898); H10F (39/182-1825); H10F (39/184-1847); H10F (39/186-1865); H10F (39/189-1898); H10F (71/1278); H10F (77/1246); H10F (77/146); H10F (77/206); H10F (30/223); H10F (39/103); G01S (7/4863); G01S (17/10); G01S (17/894); H10K (30/88); H10K (30/87); H10N (30/00); H10N (39/00); H10N (30/30-308); H10N (30/40); H10N (30/20-208); H10D (8/60); H10D (8/051); H10D (8/50); H10D (62/8325); H10D (62/8503); H10D (64/01); H10D (64/64); H10D (62/126); H10D (62/60); H10D (62/106); H10D (62/102); H10D (62/124); H10D (30/015); H10D (30/475); H10D (30/6737); H10D (30/6738); H10D (30/675); H10D (62/85); H10D (64/258); H10D (62/149); H10D (62/343); H10D (64/256); H10D (84/811); H10D (84/01); H10D (84/05); H10D (84/40); H01L (25/167); H03H (3/08); H03H (9/02976); H03H (2001/0064)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7112860	12/2005	Saxler	N/A	N/A
7294540	12/2006	Lee	N/A	N/A
8304271	12/2011	Huang	N/A	N/A
2008/0169474	12/2007	Sheppard	257/E27.014	H10N 39/00

Primary Examiner: Rahman; Moin M

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an integrated fabricating method of a structure with a photodiode, a high electron mobility transistor (HEMT) and a surface acoustic wave (SAW) device.

2. Description of the Prior Art

(2) Photodiodes are light detectors that can convert light into current or voltage signals. They can output corresponding analog electrical signals or be used in controlling switches and processing digital signals based on the degree of light received. An HEMT is a kind of field effect transistor, which uses two materials with different energy gaps to form a heterojunction to provide a channel for carriers. Compared with traditional transistors, the HEMT has higher electron speed and electron density, so it can be used to increase speed of switch on or off. A surface acoustic wave filter (SAW Filter) can be widely used in various wireless communication systems, TV sets, VCRs and GPS receivers. Its main function is to filter out noise. The SAW Filter is easier to install than traditional filters.

(3) However, with the shrinking demand for semiconductor chips, integration of circuits can increase component integration and reduce process steps. In addition, because the size of the chip becomes smaller, the distance of electron movement is reduced, so the speed of the chip can also be increased.

SUMMARY OF THE INVENTION

(4) In view of this, the present invention provides an integrated process and structure having a photodiode, an HEMT and an SAW device to decrease fabricating cost and integrate devices on a chip.

(5) According to a preferred embodiment of the present invention, a photodiode, an HEMT and an SAW device includes a photodiode including a first III-V semiconductor layer, a first N-type III-V semiconductor layer, a multi-quantum well (MQW) layer, a second III-V semiconductor layer, a P-type III-V semiconductor layer and a first electrode disposed from bottom to top. A second electrode is disposed on and contacts the first N-type III-V semiconductor layer. An HEMT includes a channel layer. An active layer is disposed on the channel layer. A P-type gate is disposed on the active layer. A gate electrode is disposed on and contacts the P-type gate. Two source/drain electrodes are respectively disposed at two sides of the P-type gate, wherein the two source/drain electrodes are embedded within the active layer and the channel layer. Schottky contact is between the first electrode and the P-type III-V semiconductor layer, and between the gate electrode and the P-type gate, Ohmic contact is between the second electrode and the first N-type III-V semiconductor layer, between one of the two source/drain electrodes and the active layer and between the other one of two source/drain electrodes and the active layer.

(6) According to another preferred embodiment of the present invention, a fabricating method of a structure with a photodiode, an HEMT and an SAW device includes providing a substrate which is divided into a photodiode region and a transistor region. Later, a first III-V semiconductor layer is formed to cover the substrate. Next, two first recesses is formed to be embedded into the first III-V semiconductor layer within the transistor region. Subsequently, an N-type III-V semiconductor layer and an MQW layer are formed to cover the first III-V semiconductor layer from bottom to top, wherein the N-type III-V semiconductor layer fills up the two first recesses. After that, the N-type III-V semiconductor layer on a top surface of the first III-V semiconductor layer and the MQW layer on the top surface of the first III-V semiconductor layer are removed. After removing the N-type III-V semiconductor layer and the MQW layer on the top surface of the first III-V semiconductor layer, a second III-V semiconductor layer and a P-type III-V semiconductor material layer are formed from bottom to top to stack within the photodiode region and the transistor region. Next, part of the P-type III-V semiconductor material layer is removed, wherein the P-type III-V semiconductor material layer remaining within the photodiode region becomes a

P-type semiconductor layer, and the P-type III-V semiconductor material layer remaining within the transistor region becomes a P-type gate. Later, two first conductive layers are respectively formed to cover the P-type III-V semiconductor material layer and the P-type gate. Then, two second recesses are respectively formed in each of the first recesses. Finally, three second conductive layers are formed, wherein two of the three second conductive layers respectively fill in the two second recesses, one of the three second conductive layers contacts the N-type III-V semiconductor layer within the photodiode region.
(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** to FIG. **10** depict a fabricating method of a structure with a photodiode, an HEMT and an SAW device according to a preferred embodiment of the present invention, wherein:
(2) FIG. **1** depicts a substrate divided into a photodiode region, a transistor region and an SAW region;
(3) FIG. **2** depicts a fabricating stage following FIG. **1**;
(4) FIG. **3** depicts a fabricating stage following FIG. **2**;
(5) FIG. **4** depicts a fabricating stage following FIG. **3**;
(6) FIG. **5** depicts a fabricating stage following FIG. **4**;
(7) FIG. **6** depicts a fabricating stage following FIG. **5**;
(8) FIG. **7** depicts a fabricating stage following FIG. **6**;
(9) FIG. **8** depicts a fabricating stage following FIG. **7**;
(10) FIG. **9** depicts a fabricating stage following FIG. **8**; and
(11) FIG. **10** depicts a fabricating stage following FIG. **9**.
DETAILED DESCRIPTION
(12) FIG. **1** to FIG. **10** depict a fabricating method of a structure with a photodiode, a high electron mobility transistor (HEMT) and a surface acoustic wave (SAW) device according to a preferred embodiment of the present invention.
(13) As shown in FIG. **1**, a substrate **10** is provided. The substrate **10** can be a silicon substrate, a germanium substrate, a gallium arsenide substrate, a silicon germanium substrate, an indium phosphide substrate, a gallium nitride substrate, a silicon carbide substrate or a silicon-on-insulator. The substrate **10** is divided into a photodiode region A, a transistor region B and an SAW region C. Next, a first III-V semiconductor layer **12***a* is formed to cover the photodiode region A, the transistor region B and the SAW region C on the substrate **10**. Later, two first recesses **14***a* are formed to embed into the first III-V semiconductor layer **12***a* within the transistor region B. The bottom of each of the first recesses **14***a* doesn't penetrate the first III-V semiconductor layer **12***a*. As shown in FIG. **2**, an N-type III-V semiconductor layer **12***n* and a multi-quantum well (MQW) layer **16** are formed to cover the first III-V semiconductor layer **12***a* from bottom to top. The N-type III-V semiconductor layer **12***n* fills up the first recesses **14***a*. The MQW layer **16** preferably includes a stack layer made of gallium nitride and indium gallium nitride. The top surface of the N-type III-V semiconductor layer **12***n* and the top surface of the MQW layer **16** are parallel to the top surface of the substrate **10**.
(14) As show in FIG. **3**, both of the N-type III-V semiconductor layer **12***n* and the MQW layer **16** on the top surface of the first III-V semiconductor layer **12***a* within the transistor region B and the SAW region C are removed. That is, only the N-type III-V semiconductor layer **12***n* and the MQW layer **16** within the photodiode region A and the N-type III-V semiconductor layer **12***n* within the

first recesses **14***a* remain. As shown in FIG. **4**, a second III-V semiconductor layer **12***b* and a P-type III-V semiconductor material layer **12***p* are formed from bottom to top to stack within the photodiode region A, the transistor region B and the SAW region C. In details, the second III-V semiconductor layer **12***b* within the photodiode region A contacts the MQW layer **16**. The second III-V semiconductor layer **12***b* within the transistor region B contacts the first III-V semiconductor layer **12***a* and the N-type III-V semiconductor layer **12***n* within the first recesses **14***a*. The second III-V semiconductor layer **12***b* within the SAW region C contacts the first III-V semiconductor layer **12***a*. Moreover, the first III-V semiconductor layer **12***a*, the N-type III-V semiconductor layer **12***n*, the MQW layer **16**, the second III-V semiconductor layer **12***b* and the P-type III-V semiconductor material layer **12***p* are preferably formed by epitaxial processes.

(15) As shown in FIG. **5**, the P-type III-V semiconductor material layer **12***p* is patterned to remove part of the P-type III-V semiconductor material layer **12***p*. In this way, the P-type III-V semiconductor material layer **12***p* remaining within the photodiode region A becomes a P-type semiconductor layer **12***p***1**, and the P-type III-V semiconductor material layer **12***p* remaining within the transistor region B becomes a P-type gate **12***p***2**. The P-type III-V semiconductor material layer **12***p* within the SAW region C is entirely removed. The P-type semiconductor layer **12***p***1** only covers part of the second III-V semiconductor layer **12***b* within the photodiode region A. The P-type gate **12***p***2** is between the two first recesses **14***a*. Next, a first conductive material layer **18** is formed to conformally cover the P-type semiconductor layer **12***p***1**, the P-type gate **12***p***2** and the second III-V semiconductor layer **12***b*. The first conductive material layer **18** is preferably formed by a deposition process. The first conductive material layer **18** is preferably Schottky metal and the Schottky metal includes a stacked layer formed by TiN/Al/TiN, Ni/Au, W/Au or Ni/Ag.

(16) As shown in FIG. **6**, the first conductive material layer **18** is patterned to separate the first conductive material layer **18** into three first conductive layers **18***a*/**18***b*/**18***c*. The first conductive material layer **18** can be patterned by an etching process. Later, the first III-V semiconductor layer **12***a* and the second III-V semiconductor layer **12***b* are etched to form two recesses **20** near an edge of the transistor region B. Then, an insulating layer **22** fills into the recesses **20** to isolate the transistor region B from other regions. The first III-V semiconductor layer **12***a* within the transistor region B will serve as a channel layer for an HEMT afterwards. The second III-V semiconductor layer **12***b* within the transistor region B will serve as an active layer for the HEMT afterwards. Moreover, three first conductive layer **18***a*/**18***b*/**18***c* respectively cover and contact the P-type III-V semiconductor layer **12***p***1**, the P-type gate **12***p***2** and the second III-V semiconductor layer **12***b* within the SAW region C. The first conductive layer **18***a* contacting the P-type III-V semiconductor layer **12***p***1** within the photodiode region A will serve as a first electrode for a photodiode. The first conductive layer **18***b* contacting the P-type gate **12***p***2** within the transistor region B will serve as a gate electrode for the HEMT. The first conductive layer **18***c* contacting the second III-V semiconductor layer **12***b* within the SAW region C will serve as an electrode for an SAW device.

(17) As shown in FIG. **7**, a protective layer **24** is formed to conformally cover the three first conductive layers **18***a*/**18***b*/**18***c*, the N-type III-V semiconductor layer **12***n* and the second III-V semiconductor layer **12***b*. Next, the protective layer **24**, the second III-V semiconductor layer **12***b* and the MQW layer **16** within the photodiode region A are etched and the second III-V semiconductor layer **12***b* and the first III-V semiconductor layer **12***a* within the transistor region B are etched. In this way, the second III-V semiconductor layer **12***b* and the MQW layer **16** not covered by the first conductive layer **18***a* within the photodiode region A are removed and two second recesses **14***b* are respectively formed within the first recesses **14***a*. The second recesses **14***b* extend into the second III-V semiconductor layer **12***b* and the protective layer **24**. The first conductive layer **18***c* and the protective layer **24** within the SAW region C are not etched. Now, part of the N-type III-V semiconductor layer **12***n* within the photodiode region A is exposed, and the N-type III-V semiconductor layer **12***n* remains around the bottom of the second recesses **14***b*.

(18) As shown in FIG. **8**, three second conductive layers **26***a*/**26***b*/**26***c* are formed. Two of the

second conductive layers **26***b*/**26***c* respectively fill in the two second recesses **14***b*, one of the second conductive layers **26***a* contacts the N-type III-V semiconductor layer **12***n* within the photodiode region A. The second conductive layers **26***a*/**26***b*/**26***c* are ohmic metal, and the ohmic metal includes a stacked layer of Ti/Al/TiN, Si/Ti/Al/TiN, Si/Ti/Ta/Al/TiN or Si/Ti/Al/Ti/Au. The second conductive layers **26***b*/**26***c* within the second recesses **14***b* will serve as source/drain electrodes of the HEMT. The second conductive layer **26***a* within the photodiode region A will serve as a second electrode for the photodiode.

(19) As shown in FIG. **9**, a dielectric layer **28** is formed to cover the photodiode region A, the transistor region B and the SAW region C entirely. Next, numerous contact plugs **30** are formed to penetrate the dielectric layer **28**. The contact plugs **30** respectively contact the first conductive layer **18***a* and the second conductive layer **26***a* within the photodiode region A, and contact the second conductive layers **26***b*/**26***c* within the transistor region B and the first conductive layer **18***c* within the SAW region C. As shown in FIG. **10**, part of the dielectric layer **28** and part of the protective layer **24** within the photodiode region A and the SAW region C are removed to expose part of the first conductive layer **18***a* within the photodiode region A and expose part of the first conductive layer **18***c* within the SAW region C. Next, the first conductive layer **18***c* within the SAW region C is etched to form an interdigital transducer **18***c***1**. Now, a structure **100** with a photodiode, an HEMT and an SAW device of the present invention is completed.

(20) As shown in FIG. **10**, the structure **100** with a photodiode, an HEMT and an SAW device includes a photodiode P, an HEMT H and an SAW device S. The photodiode P includes a first III-V semiconductor layer **12***a*, a first N-type III-V semiconductor layer **12***n*, a MQW layer **16**, a second III-V semiconductor layer **12***b*, a P-type III-V semiconductor layer **12***p***1** and a first electrode (first conductive layer **18***a*) disposed from bottom to top. A second electrode (second conductive layer **26***a*) is disposed on and contacts the first N-type III-V semiconductor layer **12***n*. The HEMT H includes a channel layer (first III-V semiconductor layer **12***a*) and an active layer (second III-V semiconductor layer **12***b*) disposed on the channel layer. A P-type gate **12***p***2** is disposed on the active layer. A gate electrode (first conductive layer **18***b*) is disposed on and contacts the P-type gate **12***p***2**. Two source/drain electrodes (second conductive layers **26***b*/**26***c*) are respectively disposed at two sides of the P-type gate **12***p***2**. The two source/drain electrodes are embedded within the active layer and the channel layer. The two N-type III-V semiconductor layers **12***n* are embedded within the channel layer. Each of the N-type III-V semiconductor layers **12***n* respectively contacts one of the source/drain electrodes (second conductive layers **26***b*/**26***c*).

(21) The SAW device S includes a piezoelectric layer **30**. The piezoelectric layer **30** is formed by the first III-V semiconductor layer **12***a* and the second III-V semiconductor layer **12***b*. An interdigital transducer **18***c***1** is disposed on and contacts the piezoelectric layer **30**. It is noteworthy that Schottky contact is between the first electrode (first conductive layer **18***a*) and the P-type III-V semiconductor layer **12***p***1**, between the gate electrode (first conductive layer **18***b*) and the P-type gate **12***p***2**, and between the interdigital transducer **18***c***1** and the piezoelectric layer **30**. Ohmic contact is between the second electrode (second conductive layer **26***a*) and the N-type III-V semiconductor layer **12***n*, between the source/drain electrode (second conductive layers **26***b*) and the active layer (second III-V semiconductor layer **12***b*) and between the source/drain electrode (second conductive layers **26***c*) and the active layer.

(22) Moreover, the N-type III-V semiconductor layer **12***n* of the photodiode P and N-type III-V semiconductor layer **12***n* of the HEMT H are originally formed by the same material layer. Therefore, the N-type III-V semiconductor layer **12***n* of the photodiode P and N-type III-V semiconductor layer **12***n* of the HEMT H are of the same material. According to a prefer embodiment of the present invention, the N-type III-V semiconductor layer **12***n* is N-type gallium nitride. Furthermore, the P-type III-V semiconductor layer **12***p***1** and the P-type gate **12***p***2** are made of the P-type III-V semiconductor material layer **12***p*, therefore, the P-type III-V semiconductor layer **12***p***1** and the P-type gate **12***p***2** are of the same material. According to a prefer embodiment of

the present invention, the P-type III-V semiconductor layer **12***p***1** and the P-type gate **12***p***2** are P-type gallium nitride. The source/drain electrodes (second conductive layers **26***b*/**26***c*) and the second electrode (second conductive layers **26***a*) are all second conductive layers, therefore they have the same material. According to a prefer embodiment of the present invention, the source/drain electrodes and the second electrode are made of ohmic metal, and the ohmic metal includes a stacked layer of Ti/Al/TiN, Si/Ti/Al/TiN, Si/Ti/Ta/Al/TiN or Si/Ti/Al/Ti/Au. The first electrode (first conductive layer **18***a*), the gate electrode (first conductive layer **18***b*), the interdigital transducer **18***c***1** are all made of the first conductive material layer **18**. Therefore, the first electrode, the gate electrode and the interdigital transducer **18***c***1** are all made of the same material. According to a prefer embodiment of the present invention, the first electrode, the gate electrode, the interdigital transducer **18***c***1** are made of Schottky metal and the Schottky metal includes a stacked layer of TiN/Al/TiN, Ni/Au, W/Au or Ni/Ag.
(23) Moreover, the first III-V semiconductor layer **12***a* is gallium nitride. The second III-V semiconductor layer **12***b* is aluminum gallium nitride. The MQW layer **16** preferably includes a stacked layer made of gallium nitride and indium gallium nitride.
(24) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A structure with a photodiode, a high electron mobility transistor (HEMT) and a surface acoustic wave (SAW) device, comprising: a photodiode, comprising: a first III-V semiconductor layer; a first N-type III-V semiconductor layer disposed on and contacting the first III-V semiconductor layer; a multi-quantum well (MQW) layer disposed on and contacting the first N-type III-V semiconductor layer; a second III-V semiconductor layer disposed on and contacting the MQW layer; a P-type III-V semiconductor layer disposed on and contacting the second III-V semiconductor layer; a first electrode disposed on and contacting the P-type III-V semiconductor layer; and a second electrode disposed on and contacting the first N-type III-V semiconductor layer; an HEMT comprising: a channel layer; an active layer disposed on the channel layer; a P-type gate disposed on the active layer; a gate electrode disposed on and contacting the P-type gate; and two source/drain electrodes respectively disposed at two sides of the P-type gate, wherein the two source/drain electrodes are embedded within the active layer and the channel layer; wherein Schottky contact is between the first electrode and the P-type III-V semiconductor layer, and between the gate electrode and the P-type gate, Ohmic contact is between the second electrode and the first N-type III-V semiconductor layer, between one of the two source/drain electrodes and the active layer and between the other one of two source/drain electrodes and the active layer.
2. The structure with a photodiode, an HEMT and an SAW device of claim 1, further comprising: an SAW device, wherein the SAW device comprises: a piezoelectric layer; an interdigital transducer disposed on and contacting the piezoelectric layer; wherein Schottky contact is between the interdigital transducer and the piezoelectric layer.
3. The structure with a photodiode, an HEMT and an SAW device of claim 1, further comprising two second N-type III-V semiconductor layers embedded within the channel layer, and each of the two second N-type III-V semiconductor layers respectively contacting one of the two source/drain electrodes.
4. The structure with a photodiode, an HEMT and an SAW device of claim 3, wherein the first III-V semiconductor layer and the two second N-type III-V semiconductor layers are made of the same material.
5. The structure with a photodiode, an HEMT and an SAW device of claim 4, wherein the first III-

V semiconductor layer and the two second N-type III-V semiconductor layers are made of N-type gallium nitride.

6. The structure with a photodiode, an HEMT and an SAW device of claim 1, wherein the P-type III-V semiconductor layer and the P-type gate are made of the same material.

7. The structure with a photodiode, an HEMT and an SAW device of claim 6, wherein the P-type III-V semiconductor layer and the P-type gate are made of P-type gallium nitride.

8. The structure with a photodiode, an HEMT and an SAW device of claim 1, wherein the two source/drain electrodes and the second electrode are made of the same material.

9. The structure with a photodiode, an HEMT and an SAW device of claim 8, wherein the two source/drain electrodes and the second electrode are made of ohmic metal, and the ohmic metal comprises Ti/Al/TiN, Si/Ti/Al/TiN, Si/Ti/Ta/Al/TiN or Si/Ti/Al/Ti/Au.

10. The structure with a photodiode, an HEMT and an SAW device of claim 2, wherein the first electrode, the gate electrode and the interdigital transducer are made of the same material.

11. The structure with a photodiode, an HEMT and an SAW device of claim 2, wherein the first electrode, the gate electrode and the interdigital transducer are made of Schottky metal and the Schottky metal comprises TIN/Al/TIN, Ni/Au, W/Au or Ni/Ag.